

PRODUCT REQUIREMENTS DOCUMENT

EdTech B2B2C Platform

Minimum Viable Product — full-scope specification for CRM, learning delivery, assessments, payments, commission settlement, and the student & parent mobile app.

DOCUMENT VERSION 1.0 (Draft for build)	DATE 10 September 2026
STATUS Approved scope — ready for estimation	OWNER Founding team
AUDIENCE External engineering partner (Toptal), architects, QA	BUSINESS MODEL B2B2C — commission contracts with institutes
DELIVERY TARGETS Web CRM, Marketing site, NestJS API, React Native app	CONTENT SCOPE V1 Classes 9–12, all subjects (arch. supports 6–12)

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1. Executive Summary & Business Model

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The platform is an online learning business for secondary and senior-secondary students (Classes 9–12 at launch), sold through a **B2B2C** channel: the company signs commission contracts with schools and colleges (“Institutes”), and those Institutes enroll their students onto the platform.

1.1 How the business works

- The company produces and curates **video lessons, PDF study material, and test series** for every subject of Classes 9, 10, 11 and 12.
- Teaching and grading are delivered by the company’s **own central pool of teachers**, serving every Institute. Institutes do not supply teaching staff.
- Each Institute signs a **written, e-signed contract** committing to a **minimum number of students** over a defined period, in exchange for a **commission** that is either a **percentage of student fees** or a **fixed amount per student**.
- When a student pays, the platform automatically **splits and settles the Institute’s commission** through Razorpay Route, and tracks the Institute’s progress against its committed student count.
- Students can also arrive **directly from the marketing website**; these are attributed to a system “Direct” tenant so a single commission code-path applies everywhere.

1.2 What we are building (four deployables)

#	Deployable	Stack	Primary users
1	Marketing Website	Next.js, server-side rendered , separate codebase & deploy	Public visitors, institute & student lead capture, SEO
2	CRM + Admin Web App	React / Next.js, hosted on a subdomain (e.g. app.domain.com)	Super Admin, Ops, Sales, Marketing, Teachers, Institute Admins — and Students & Parents (full learning experience, web)
3	Backend API	Node.js + NestJS + MongoDB , REST, API-first	All clients
4	Student & Parent Mobile App	React Native (iOS + Android)	Students, Parents — same feature set as the web learning experience

KEY DECISION

Students and Parents use **both** the web CRM app and the React Native app, with **feature parity**: dashboard, videos, study material, test series, proctored subjective tests, and AI test analysis are available on both. The backend is a single API serving every client.

1.3 MVP success criteria

- An Institute can be onboarded, sign a contract, and have its students enrolled and paying within one working day.
- A student can watch lessons, download material, take an MCQ test, receive AI-generated weak-topic analysis, and be served an adaptive retest — end to end — on web and mobile.
- A subjective test can be taken under proctoring, graded by a teacher, and returned with analysis.

- Every rupee collected is reconciled: platform revenue, tax, coupon cost, and Institute commission settled and reportable.

- Super Admin can impersonate any user for support, with a full audit trail.

2. Goals, Non-Goals & MVP Boundary

2.1 Goals

- One backend, four clients, clean REST API, multi-tenant by Institute.
- Complete contract-to-commission lifecycle with automated settlement.
- Complete learn-to-assess-to-improve loop with AI analysis and adaptive retests.
- Anti-piracy content delivery (signed URLs, per-user watermarking, screen-capture protection on mobile).
- Role-based access with impersonation and full audit logging.
- Compliant handling of minors' data and proctoring recordings (India DPDP Act).

2.2 Non-Goals for v1

- Live / synchronous online classes (Zoom / Meet integration) — planned v2.
- Regional-language content and UI — English only at launch, i18n-ready structure.
- SMS / WhatsApp notifications — abstraction in place, provider added later.
- Public marketplace / self-serve Institute signup without a signed contract.
- Advanced gamification, social feed, peer messaging.
- Offline-first web app; only the mobile app supports offline content.

2.3 Phase tags used in this document

P1 — MVP must ship at launch

P1.1 — FAST FOLLOW within ~6–8 weeks of launch

P2 later

3. Personas, Roles & Permissions Matrix

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3.1 Roles

Role	Platform	Scope	Purpose
Super Admin	Web	Global	Full access, platform configuration, billing, impersonate any user , view audit log.
Sub-Admin / Ops	Web	Global (restricted)	Day-to-day operations; no destructive config, no payout release, no role management.
Sales (internal)	Web	Assigned institutes / leads	Lead pipeline, institute onboarding, contract drafting, follow-ups.
Marketing (internal)	Web	Global marketing objects	Campaigns, UTM, coupons , Brevo broadcasts, website leads, acquisition analytics. Separate role from Sales.
Teacher / Content Manager (internal)	Web	Platform-wide central pool	Upload & publish content, build question bank, grade subjective tests, review proctoring flags.
Institute Admin	Web	Own Institute (tenant)	Bulk-import & manage own students, view own dashboard, commitment progress, commission earned/pending, own referral code, own coupons if enabled.
Student	Web + Mobile	Own account	Full learning experience: dashboard, videos, material, tests, proctored tests, analysis.
Parent	Web + Mobile	Linked student(s)	Read-only progress & results view; consent / approval for minor signup and proctoring.

REMOVED FROM EARLIER DRAFTS

The “Institute Staff / Coordinator” role is removed. Student onboarding and bulk import sit with **Institute Admin** only, because teaching is delivered centrally.

3.2 Permissions matrix (excerpt)

✓ = allowed · = not allowed R = read-only O = own records only

Capability	Super Admin	Ops	Sales	Mktg	Teacher	Inst. Admin	Student	Parent
Impersonate user	✓	·	·	·	·	·	·	·
Manage roles & permissions	✓	·	·	·	·	·	·	·
Create / edit Institute & contract	✓	✓	✓ draft	·	·	·	·	·
Approve contract / release payout	✓	·	·	·	·	·	·	·
Manage coupons & campaigns	✓	✓	·	✓	·	R own	·	·

Capability	Super Admin	Ops	Sales	Mktg	Teacher	Inst. Admin	Student	Parent
Upload & publish content	✓	R	•	•	✓	•	•	•
Build question bank / tests	✓	R	•	•	✓	•	•	•
Grade subjective tests	✓	•	•	•	✓	•	•	•
Review proctoring flags	✓	✓	•	•	✓	•	•	•
Bulk-import students	✓	✓	•	•	•	✓ O	•	•
View Institute dashboard	✓ all	✓ all	R assigned	•	•	✓ O	•	•
Take tests / watch content	•	•	•	•	•	•	✓ O	•
View student results & analysis	✓	✓	•	•	✓	R O	✓ O	R linked
Give proctoring / signup consent	•	•	•	•	•	•	•	✓ linked

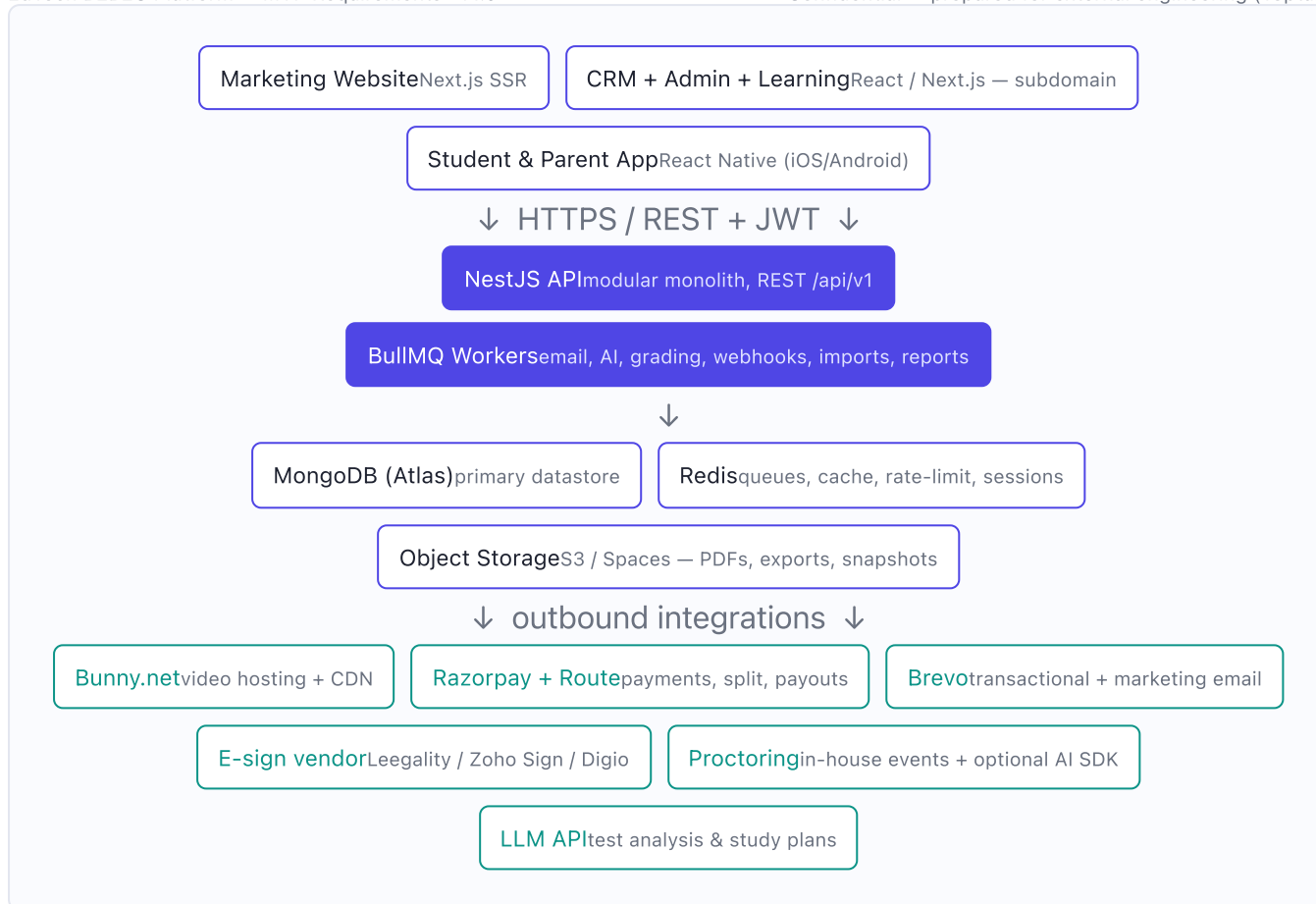
FR-RBAC-01 All authorization is enforced server-side per request; the client role only shapes navigation.

FR-RBAC-02 Every privileged or financial action writes an immutable audit entry (actor, real actor if impersonating, target, before/after, IP, timestamp).

4. System Architecture & Deployables

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4.1 Architectural principles

- **Modular monolith** NestJS API (modules: auth, tenants, institutes, contracts, crm, marketing, students, catalog, media, assessments, ai, proctoring, payments, commission, notifications, reporting, admin). Split to services only if load demands it.
- **API-first & versioned** (/api/v1). All four clients consume the same contract; the mobile app must never require a bespoke endpoint shape.
- **Multi-tenant** by `instituteId` (with a reserved `DIRECT` tenant). Tenant scoping applied in a global query guard; cross-tenant access is a test-covered invariant.
- **Async by default** for anything slow or external — queued via BullMQ with retry/backoff and dead-letter handling.
- **Stateless API nodes**; sessions/refresh tokens and rate-limit counters in Redis; horizontal scale behind a load balancer.
- **Signed, expiring URLs** for every piece of protected media (video playback tokens, PDF links, proctoring uploads).
- **Idempotent webhooks** (Razorpay, Bunny, e-sign) keyed by event id; all webhook signatures verified.

4.2 Environments

Local	Docker Compose: API, Mongo, Redis, MinIO (S3-compatible), Mailhog.
Staging	Full parity, sandbox keys for all integrations, seed data, open to the client for UAT.
Production	Isolated project/VPC, managed Mongo + Redis, WAF, backups, monitoring.

5. Technology Stack

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Layer	Choice	Notes
Marketing website	Next.js (App Router), SSR/SSG, TypeScript	Rendered server-side, not client-side. Deployed as a Node server (not static export) so pages are server-rendered on request.
CRM / Admin / Learning web	React 18 + Next.js or Vite SPA, TypeScript, component library (MUI / shadcn), TanStack Query	Runs on a subdomain. Client-rendered app behind auth.
Mobile app	React Native (bare or Expo with dev-client), TypeScript	iOS + Android. Native modules for screen-capture protection, secure storage, background/app-state detection, video DRM playback.
API	Node.js LTS, NestJS, TypeScript	REST, OpenAPI/Swagger generated, class-validator DTOs, Passport JWT.
Database	MongoDB (Atlas), Mongoose ODM	Replica set, daily backups, PITR on production.
Cache / queue	Redis + BullMQ	Job queues, caching, rate limiting, refresh-token store.
Object storage	AWS S3 or DigitalOcean Spaces	PDFs, generated reports/invoices, proctoring snapshots, CSV import artifacts.
Video	Bunny.net Stream + CDN	Signed playback, watermarking, HLS.
Payments	Razorpay + Razorpay Route	Orders, subscriptions, refunds, linked accounts, transfers/settlements.
Email	Brevo (transactional API + marketing)	Templates, lists, automation.
E-signature	Leegality (recommended) / Zoho Sign / Digio	Aadhaar / OTP e-sign, India-compliant audit trail.
AI analysis	Managed LLM API + deterministic scoring layer	Rules compute weak topics; LLM writes the narrative feedback & study plan.
Push notifications	Firebase Cloud Messaging (Android + iOS via APNs)	Mobile only.
Observability	Sentry, structured logs, uptime monitor, health checks	Error + performance monitoring across API, web, mobile.
CI/CD	GitHub Actions	Lint, test, build, deploy; EAS build/submit for mobile.
IaC	Terraform	Reproducible infra for staging + production.
Hosting	AWS recommended (ECS Fargate + ALB); DigitalOcean App Platform acceptable for lower ops	See §12. Mongo & Redis managed in both.

6. Functional Requirements

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Each sub-section lists requirements with stable IDs and, for flagship features, acceptance criteria.

Unless tagged otherwise every requirement is **P1 — MVP**.

6.1 Identity, Authentication & RBAC

- **FR-AUTH-01** Email + password registration and login; passwords hashed with bcrypt/argon2; email verification required before first paid action.
- **FR-AUTH-02** OTP login (email) for students/parents; optional biometric unlock on mobile after first login.
- **FR-AUTH-03** JWT access token (short-lived) + rotating refresh token stored server-side (Redis); refresh revocation on logout / password change.
- **FR-AUTH-04** Forgot-password with time-boxed single-use token.
- **FR-AUTH-05** TOTP 2FA: **mandatory** for Super Admin, Ops, Sales, Marketing, Teacher; optional for Institute Admin.
- **FR-AUTH-06** **Device / session limit** for Student accounts (default 2 active devices); registering a 3rd forces logout of the oldest; visible session list.
- **FR-AUTH-07** **Impersonation**: Super Admin starts "View as <user>"; a persistent banner shows on every screen with an **Exit impersonation** action; impersonated session cannot change passwords, 2FA, or payout settings.
- **FR-AUTH-08** Parent–student linking via invite code or Institute-provided mapping; a Parent may link multiple students; a student may have one primary Parent.
- **FR-AUTH-09** Global **audit log** with filter/search/export; entries immutable; retained per §11 retention policy.
- **FR-AUTH-10** Account logout / exponential backoff after repeated failed logins; suspicious-login email.

ACCEPTANCE — IMPERSONATION

Given a Super Admin impersonating a Student, when they open any page or call any API, then responses reflect the Student's permissions and data, the audit log records both the real admin id and the target id, and ending impersonation returns the admin to their own session without re-login.

6.2 Institute & Partner Management

- **FR-INST-01** Create Institute: legal name, display name, type (school/college/coaching), address, city/state, GSTIN, PAN, primary + billing contacts, logo.
- **FR-INST-02** KYC document upload (registration certificate, PAN, GST, cancelled cheque / bank proof) with per-document verification status.
- **FR-INST-03** Institute lifecycle: PROSPECT → ONBOARDING → ACTIVE → SUSPENDED → CHURNED; state changes are audited and can gate student enrollment.
- **FR-INST-04** Unique **Institute referral code** auto-generated; used at student signup for commission attribution; regeneratable with history.
- **FR-INST-05** Razorpay Route **Linked Account** creation & KYC status tracking per Institute; commission settlement blocked until active.
- **FR-INST-06** **Commitment tracking**: committed student count vs enrolled vs paid; progress %, projected shortfall, and alerts at configurable thresholds (e.g. 60 days to term-end and <70% of commitment).

- **FR-INST-07** Commission ledger per Institute: accrued, pending settlement, settled, adjustments, disputes, running balance.

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- **FR-INST-08** Institute data export (own students + performance) as CSV/Excel.
- **FR-INST-09** Notes, tasks, and file attachments on the Institute record (shared with Sales CRM §6.4).

6.3 Contract Handling & E-Signature

- **FR-CTR-01** Contract created from a versioned template or by uploading a prepared PDF.
- **FR-CTR-02** Structured terms captured alongside the document:
 - Commission model: PERCENTAGE (of net student fee) or FIXED (₹ per enrolled/paid student).
 - Commission rate / amount; optional **slabs** (e.g. 10% up to 200 students, 12% beyond).
 - **Minimum student commitment** (count) and commitment period (start & end dates).
 - Settlement cycle: PER_TRANSACTION or WEEKLY or MONTHLY.
 - Commission basis: gross vs **net of coupon** (default net); refund clawback rule.
 - Renewal terms, notice period, termination clause.
- **FR-CTR-03** E-signature via integrated vendor: send for signature, track signer status, receive signed PDF + tamper-proof audit trail via webhook.
- **FR-CTR-04** Contract states: DRAFT → SENT → SIGNED → ACTIVE → EXPIRING → EXPIRED / RENEWED / TERMINATED.
- **FR-CTR-05** Only an **ACTIVE** contract enables commission calculation; term changes require a new version or addendum (old versions retained & viewable).
- **FR-CTR-06** Automatic **expiry reminders** to Sales + Super Admin at 60/30/7 days.
- **FR-CTR-07** Contract approval gate: Super Admin countersign / approve before **ACTIVE**.

ACCEPTANCE — CONTRACT TO COMMISSION

Given an ACTIVE percentage contract at 12% net, when a mapped student pays ₹10,000 with a ₹1,000 coupon, then commission accrued = $12\% \times ₹9,000 = ₹1,080$, a Route transfer of ₹1,080 is scheduled per the settlement cycle, the Institute ledger updates, and the student counts toward commitment on successful payment capture.

6.4 Sales CRM

- **FR-CRM-01** Lead entity (Institute prospect): source, owner, stage, region, expected student count, expected annualised value, next action date.
- **FR-CRM-02** Configurable **pipeline / kanban**; drag between stages; win/loss reasons on close.
- **FR-CRM-03** Activities: tasks, reminders, call logs, meeting notes, attachments; activity timeline per lead/Institute.
- **FR-CRM-04** Assignment & ownership transfer; “my leads” vs team view; territory/region tag.
- **FR-CRM-05** Send email from CRM via Brevo using templates; log opens/clicks on the timeline.
- **FR-CRM-06** Website “Partner with us” form creates a Lead automatically with source = website.
- **FR-CRM-07** Convert Lead → Institute (carries over contacts, notes, owner).
- **FR-CRM-08** Reports: pipeline value by stage, stage conversion, rep activity, win rate, forecast (Σ expected students $\times$ price $\times$ commission).

6.5 Marketing, Campaigns & Coupons

- **FR-MKT-01** **Coupon engine**: code, discount type (PERCENT / FLAT), value, currency.

- **FR-MKT-02** Constraints: valid-from/valid-to, total usage cap, per-user cap, minimum order value, first-purchase-only, applicable classes / subjects / plans.

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- **FR-MKT-03** Scope: GLOBAL, INSTITUTE (auto-attributes enrollment to that Institute), CAMPAIGN.
- **FR-MKT-04** Stacking rule per coupon (allow / disallow combining with other offers); server is the single source of truth for final price.
- **FR-MKT-05** Campaign object with UTM capture; UTM + referral code persisted on the Student record at signup.
- **FR-MKT-06** Brevo broadcasts: contact lists & segments, scheduled sends, unsubscribe sync, marketing vs transactional separation.
- **FR-MKT-07** Acquisition dashboard: signups by source, coupon redemptions & discount cost, campaign ROI, funnel (visit → signup → paid).
- **FR-MKT-08** **P1.1** Referral rewards for students; A/B pricing blocks on the website.

6.6 Student Management

- **FR-STU-01** Enrollment paths: (a) Institute bulk import, (b) Institute/campaign referral link, (c) direct website signup (→ DIRECT tenant).
- **FR-STU-02** Student profile: name, DOB, contact, **board, class (6–12), stream, subjects, batch**, Institute mapping, parent link, status, acquisition metadata.
- **FR-STU-03** **Bulk CSV import** by Institute Admin: template download, row validation, dedupe, dry-run preview, error report, auto credential generation, welcome email with app links.
- **FR-STU-04** Status: INVITED → ACTIVE → SUSPENDED → EXPIRED → REFUNDED; access derived from status + purchased plan + class.
- **FR-STU-05** Content/test access is computed: a student sees only their class & subjects, plus free/sample items, plus anything their plan grants.
- **FR-STU-06** **Attribution & re-attribution policy**: first valid Institute code wins for the current term; later code changes are logged and take effect from the next billing cycle; disputes flagged for Ops.
- **FR-STU-07** Parent consent gate for students under 18: signup incomplete until Parent approves (see §11).

6.7 Course & Content Management (LMS admin)

- **FR-CNT-01** Taxonomy: Board → Class → Subject → Chapter → Topic → Lesson. Every content item and question is tagged to a **Topic** (drives AI analysis & test blueprints).
- **FR-CNT-02** Content types: video (Bunny), PDF/notes/resource (S3), external link. Each item: title, description, topic tags, difficulty, free/paid flag, order, status.
- **FR-CNT-03** Workflow: DRAFT → IN_REVIEW → PUBLISHED; schedule publish/unpublish; content versioning with rollback.
- **FR-CNT-04** **Bulk content tooling**: batch upload of videos (with metadata CSV), batch PDF upload, batch topic re-tagging. Treated as a first-class workstream given catalogue volume (all subjects × 4 classes).
- **FR-CNT-05** Content search & filters for admins; broken-link / missing-media report.
- **FR-CNT-06** Per-student **progress capture** API: video position & % watched, lesson complete, chapter complete, time-on-content; powers dashboards & AI.
- **FR-CNT-07** Student-side content features (web + mobile): resume playback, bookmarks, personal notes, rate/report a lesson.

6.8 Video Delivery (Bunny.net)

- **FR-VID-01** Server-side upload / TUS pipeline to Bunny; transcode to adaptive HLS renditions; poster/thumbnail generation.
- **FR-VID-02** Playback only via **token-signed, short-expiry URLs** issued per student per session; referrer / allowed-origin lock; no raw CDN URL exposed to the client.
- **FR-VID-03** **Dynamic watermark** overlaying student id / email / phone, position rotating periodically (anti screen-record).
- **FR-VID-04** Player features: speed control, quality selector, resume, chapters, captions/subtitles upload, keyboard shortcuts (web).
- **FR-VID-05** Mobile: DRM-capable player, **screenshot / screen-record blocking** (Android FLAG_SECURE, iOS capture detection), screen-mirroring / external-display block during playback.
- **FR-VID-06** **P1 Offline download** (mobile): encrypted local store, per-title download, device-bound, expiry (e.g. 30 days / re-validate every 7), max concurrent downloads, admin toggle per content item to allow/deny offline.
- **FR-VID-07** Bunny webhook handling for encode-complete / failure; ret/alert on failure.
- **FR-VID-08** Playback analytics event stream (start, progress pings, complete, errors) stored for reporting & AI.

6.9 Study Material / PDF

- **FR-PDF-01** Upload PDFs/resources to object storage; served only via **signed expiring links**.
- **FR-PDF-02** In-app / in-browser PDF viewer with **per-student watermark** (name, id, timestamp) stamped on view.
- **FR-PDF-03** Per-item flags: allow/deny download, allow/deny print.
- **FR-PDF-04** Mobile offline access for permitted PDFs uses the same encrypted store & expiry as video (FR-VID-06).
- **FR-PDF-05** Bookmarks & personal notes per document (student).

6.10 Assessment & Test Engine

- **FR-ASM-01** **Question bank** item types: MCQ single-correct, MCQ multiple-correct, numerical/integer, subjective/descriptive. Fields: stem (rich text + images + LaTeX), options, correct answer(s), marks, negative marks, expected time, difficulty, solution/explanation, board/class/subject/chapter/**topic** tags, status.
- **FR-ASM-02** Bulk question import (Excel/CSV/Word template) with image handling & validation report.
- **FR-ASM-03** **Test types**: topic-wise, chapter-wise, mock / full-length, **half-yearly**, **full-yearly**, and **adaptive weak-area retest** (§6.11).
- **FR-ASM-04** Test builder: fixed question list or **blueprint** (pick N of M from a pool per chapter/difficulty), section definitions, per-section timing, marking scheme, negative marking, attempts allowed, availability window, pause/resume policy.
- **FR-ASM-05** **Randomisation**: shuffle question order and option order per attempt; per-student question set from pools; anti-collusion seed per attempt.
- **FR-ASM-06** Test-taking runtime (web + mobile): countdown timer, autosave every N seconds, resume after crash/network loss, submit-on-timeout, question palette (answered / marked-for-review / unseen), section lock.

- **FR-ASM-07** Subjective answer input: rich text, LaTeX/equation, and **image upload of handwritten answer** (multi-page).

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- **FR-ASM-08** **Auto-grading** for MCQ/numerical on submit. **Manual grading** for subjective: teacher queue, per-criterion rubric, partial marks, inline annotation, optional second-review/moderation, publish results.
- **FR-ASM-09** Result object: raw & percentage score, section scores, percentile, rank (cohort-scoped), accuracy, attempted/correct/incorrect/skipped, time per question, time vs cohort.
- **FR-ASM-10** **Topic-wise & chapter-wise strength/weakness heatmap**; comparison vs previous attempts; full solution review mode.
- **FR-ASM-11** Basic anti-cheating on MCQ: response-time anomaly detection, impossible-speed flags, duplicate-attempt prevention.
- **FR-ASM-12** **P1.1** Institute-scoped leaderboards (toggleable), scheduled test series calendars.

6.11 AI Analysis & Adaptive Retests

- **FR-AI-01** After every graded test, a deterministic layer computes per-topic mastery from accuracy, difficulty, time, and trend across attempts, and classifies each topic as **strong / borderline / weak**.
- **FR-AI-02** An LLM generates a **student-facing narrative**: what went well, top 3 weak concepts, likely misconceptions, and a prioritised **study plan** with deep links to the exact lessons/PDFs for those topics.
- **FR-AI-03** **Adaptive weak-area retest**: on demand or auto-scheduled, the engine builds a new test drawn from the student's weak topics using the blueprint sampler; difficulty adjusts based on the last attempt.
- **FR-AI-04** Progress view shows weak-topic list shrinking over time; "next best action" on the dashboard.
- **FR-AI-05** Optional diagnostic test per subject at onboarding to seed the initial mastery baseline.
- **FR-AI-06** All AI output is advisory, logged, and regenerable; prompts & model version stored with each analysis for reproducibility. No student PII beyond first name sent to the model unless configured otherwise.

ACCEPTANCE — ADAPTIVE LOOP

Given a completed chapter test where topics A and C are classified weak, when the student opens "Practice weak areas", then a new test is generated containing only A/C-tagged questions not seen in the last attempt, and on completion the analysis shows updated mastery for A and C and an updated study plan.

6.12 Proctoring (subjective tests only)

- **FR-PRO-01** Proctoring is enabled **only for subjective test types** by default; a per-test flag can enable it for any test.
- **FR-PRO-02** Pre-test: identity selfie capture + stored reference photo match prompt, camera/mic permission, environment/system check, consent acknowledgement (student + Parent consent on file for minors).
- **FR-PRO-03** During test — **web**: enforced full-screen, tab-switch / window-blur detection, copy-paste & right-click block, periodic webcam snapshots (configurable interval), optional continuous webcam recording, disconnect handling with grace period.
- **FR-PRO-04** During test — **mobile**: app-backgrounding / app-switch detection with forced return, screenshot & screen-record block, no split-screen / picture-in-picture, incoming-call interruption logging, periodic front-camera snapshots.

- **FR-PR0-05** AI flags (in-house heuristics for MVP, optional vendor SDK): no face, multiple faces, face-not-matching-reference, phone/object detected, prolonged absence, background voices.
- **FR-PR0-06** Events & media stream to a **Proctor Review dashboard** (web, Teacher/Ops): per-attempt timeline, snapshot gallery, recording playback, severity score, decision = ACCEPT / FLAG / INVALIDATE + allow retake, reviewer notes.
- **FR-PR0-07** Proctoring media stored encrypted with a defined **retention window** (default 90 days) then auto-purged; access is role-limited and audited.

6.13 Payments & Billing (Razorpay)

- **FR-PAY-01** Catalogue of plans: per-subject, per-class bundle, full-year package, multi-year; **one-time** and **subscription** (monthly/annual) both supported; Institute-negotiated price overrides.
- **FR-PAY-02** Checkout: server computes final price (base – coupon + tax); creates Razorpay Order / Subscription; client uses Razorpay Checkout.
- **FR-PAY-03** **Webhooks** (signed, idempotent): payment.captured, payment.failed, order.paid, subscription.*, refund.processed, transfer.*, settlement.*.
- **FR-PAY-04** On successful capture: activate student access, record acquisition/attribution, accrue Institute commission, trigger invoice + confirmation email, count toward Institute commitment.
- **FR-PAY-05** **Refunds**: full/partial, reason codes, Ops approval workflow, GST credit note, access revocation policy, **commission clawback** from the Institute ledger.
- **FR-PAY-06** **GST-compliant invoices**: sequential numbering, company + customer details, HSN/SAC, tax breakup, downloadable PDF; stored in object storage.
- **FR-PAY-07** Failed-payment dunning: retry schedule + Brevo emails; subscription grace period & auto-suspend.
- **FR-PAY-08** Payment & reconciliation report: gross, discount, tax, net, fees, refunds, commission, platform revenue — per day / Institute / plan.

6.14 Commission Split & Settlement (Razorpay Route)

- **FR-COM-01** Each Institute maps to a Razorpay **Linked Account**; settlement is blocked until its KYC is active.
- **FR-COM-02** At payment time the engine reads the Institute's ACTIVE contract and computes commission (percentage-of-net or fixed-per-student, honouring slabs).
- **FR-COM-03** Settlement mode per contract:
 - PER_TRANSACTION — create a Route **transfer** on the payment (immediate or on-hold).
 - WEEKLY / MONTHLY — accrue in the ledger, then a scheduled job creates a consolidated transfer/payout for the period.
- **FR-COM-04** Commission basis default = **net of coupon**; refunds create negative ledger entries and reduce the next settlement (or raise a recovery task if the balance is negative).
- **FR-COM-05** Per-Institute **settlement statement** (PDF/CSV): opening balance, accruals, adjustments, transfers, TDS note, closing balance; visible on the Institute dashboard.
- **FR-COM-06** Payout release requires Super Admin approval for consolidated cycles; all transfers reconciled against Razorpay transfer.* / settlement.* webhooks.
- **FR-COM-07** Dispute flag on a ledger line freezes that amount pending Ops resolution.

6.15 Notifications & Email (Brevo) + Push

- **FR-NOT-01** Transactional email via Brevo for: welcome/credentials, email verification, OTP, payment success/failure, refund processed, enrollment confirmed, test scheduled / starting soon, result published, subjective grading complete, contract sent/signed/expiring, commission settled, commitment-shortfall alert, suspicious login.
- **FR-NOT-02** Automated sequences: student onboarding drip, inactivity re-engagement, trial-to-paid nudges.
- **FR-NOT-03** In-app notification centre (web + mobile) with read/unread & deep links.
- **FR-NOT-04** **Push notifications** (mobile, FCM/APNs): test reminders, results, new content in the student's subjects, payment/subscription alerts. Per-category opt-out.
- **FR-NOT-05** Admin-editable templates, per-event enable/disable, and a notification log.
- **FR-NOT-06** SMS/WhatsApp behind a provider interface, **not implemented in v1**.

6.16 Student & Parent Mobile App (React Native)

Feature parity with the web learning experience, plus mobile-native protections. iOS + Android, phone-first with responsive tablet layout.

STUDENT

- **FR-APP-01** Auth: credential/OTP login, biometric unlock, device-limit enforcement (FR-AUTH-06), secure token storage (Keychain / Keystore).
- **FR-APP-02** **Dashboard**: continue watching, upcoming/available tests, pending analysis, study streak, overall & subject progress, weak-topic highlights, announcements, notifications.
- **FR-APP-03** **Videos**: signed playback, dynamic watermark, speed/quality, resume, chapters, captions; screenshot/screen-record & mirroring block; **encrypted offline downloads** with expiry.
- **FR-APP-04** **Study material**: watermarked PDF viewer, allow/deny download & print, offline for permitted items, bookmarks & notes.
- **FR-APP-05** **Test series**: all test types, timer, autosave, crash/network resume, question palette, subjective input incl. handwritten-answer photo upload.
- **FR-APP-06** **Proctoring** for subjective tests: pre-test identity + consent, app-switch/background detection, snapshot capture, screen-capture block (per FR-PRO-04).
- **FR-APP-07** **Test analysis**: score/percentile/rank, topic & chapter heatmap, attempt comparison, full solutions, AI narrative + study plan with deep links into content, one-tap "practice weak areas".
- **FR-APP-08** Payments: view plans, apply coupon, pay via Razorpay (in-app), view invoices, manage subscription. (Store-billing implications to be confirmed — see §13 Q.)
- **FR-APP-09** Offline behaviour: cached content + last-synced dashboard/results readable offline; test submission & proctoring require connectivity; graceful queue-and-sync for progress events.
- **FR-APP-10** Push notification handling & deep links; in-app notification centre.
- **FR-APP-11** Accessibility (dynamic type, screen-reader labels), crash/error reporting, forced-update mechanism, staged rollout.

PARENT

- **FR-APP-20** Link to one or more students via invite code; switch between linked students.
- **FR-APP-21** Read-only dashboard: attendance/engagement, content progress, test results & analysis summaries, upcoming tests.
- **FR-APP-22** **Consent actions**: approve minor signup, approve proctoring & recording, approve data-processing terms; consent events logged with version/timestamp.

- **FR-APP-23** Notifications: result published, low activity, payment/subscription events, teacher remarks.
- **FR-APP-24** No access to test-taking, content playback controls, or financial actions beyond viewing invoices for linked students.

6.17 Reporting & Dashboards

Audience	Key views
Super Admin	Revenue (gross/net/platform), active students, new enrollments, MRR & churn, commission payable/settled, Institute performance vs commitment, content consumption, test participation & average scores, refund rate, coupon cost, proctoring flag volume.
Institute Admin	Own students, enrollments over time, commitment progress & projected shortfall, commission earned/pending, settlement statements, student performance summary, top & at-risk students.
Sales	Pipeline value & stage conversion, activity, win rate, forecast.
Marketing	Acquisition by channel, coupon/campaign performance, funnel.
Teacher	Grading queue & SLA, class/cohort performance, item analysis (most-missed questions), proctoring review queue.
Student / Parent	Progress, streak, weak topics, results & analysis (see §6.16).

- **FR-RPT-01** All list/report views exportable to CSV/Excel; key financial reports exportable to PDF.
- **FR-RPT-02** Date-range & entity filters; tenant-scoped automatically for Institute Admins.
- **FR-RPT-03** **P1.1** Scheduled email reports.

6.18 Marketing Website (Next.js, SSR)

- **FR-WEB-01** Pages: Home, About, Courses/Classes, Pricing, **For Institutes / Partner Program**, Blog, Contact, FAQ, Legal (Terms, Privacy, Refund & Cancellation, Proctoring & Consent).
- **FR-WEB-02** **Server-side rendered** pages (SSR/SSG), semantic HTML, meta/OG/Twitter tags, sitemap.xml, robots.txt, schema.org (Organization, Course, FAQ), image optimisation, Core Web Vitals budget (LCP < 2.5s).
- **FR-WEB-03** Lead capture: Institute partnership form → CRM Lead; student signup / free-trial → backend.
- **FR-WEB-04** Coupon-aware pricing display; sample/free content CTA; app-store badges.
- **FR-WEB-05** Blog via lightweight headless CMS (recommend Sanity) or MDX; static pages acceptable at launch.
- **FR-WEB-06** Analytics (GA4 or Plausible), cookie consent, WhatsApp/contact widget, responsive, WCAG 2.1 AA target.

6.19 Platform Administration & Configuration

- **FR-ADM-01** Manage taxonomy (boards, classes, subjects, chapters, topics).
- **FR-ADM-02** Manage plans, pricing, tax rates, coupons, campaigns.
- **FR-ADM-03** Feature flags & environment config; maintenance mode.
- **FR-ADM-04** Content & question review/moderation queues.
- **FR-ADM-05** Lightweight support inbox / query ticketing (email-backed) — basic in v1.
- **FR-ADM-06** Manage notification templates & rules; view notification & webhook logs.

- **FR-ADM-07** Audit-log viewer with filters & export.

7. Data Model Overview

EdTech B2B2C Platform — MVP Requirements · v1.0

Confidential — prepared for external engineering (Toptal)

Core collections and their principal relationships. All tenant-owned documents carry `instituteId` (or `DIRECT`). All documents carry `createdAt`, `updatedAt`, and `soft-delete` where relevant.

Collection	Key fields	Relationships
User	name, email, phone, passwordHash, roles[], status, twoFactor, deviceSessions[]	→ Institute (staff/admin), → StudentProfile, → ParentLink
Institute	legalName, type, gstin, pan, status, referralCode, linkedAccountId, commitmentSummary	← Users, ← Contracts, ← Students, ← CommissionLedger
Contract	version, status, commissionModel, rate, slabs[], minStudents, periodStart/End, settlementCycle, basis, documentUrl, esignRef	→ Institute
Lead	source, stage, owner, region, expectedStudents, expectedValue, activities[]	→ Institute (on conversion), → User (owner)
StudentProfile	userId, board, class, stream, subjects[], batch, status, attribution{instituteId, coupon, utm}, parentId	→ User, → Institute, → Enrollment, → Attempt
ParentLink	parentUserId, studentIds[], consents[]	→ User
Taxonomy (Board/Class/Subject/Chapter/Topic)	name, parentId, order, code	self-referential tree
ContentItem	type(video/pdf/link), title, topicIds[], difficulty, isFree, status, bunnyVideoid / storageKey, allowDownload, allowOffline	→ Topic(s)
ContentProgress	studentId, contentItemId, position, percent, completedAt, timeSpent	→ Student, → ContentItem
Question	type, stem, options[], answerKey, marks, negMarks, difficulty, topicIds[], solution, status	→ Topic(s)
Test	type, blueprint / questionIds[], sections[], timing, markingScheme, proctored, window, attemptsAllowed	→ Questions / Topics
Attempt	testId, studentId, servedQuestions[], responses[], startedAt, submittedAt, autoScore, status	→ Test, → Student, → ProctorSession, → Result
Result	score, percent, sectionScores, percentile, rank, topicMastery[], timeAnalysis	→ Attempt
AiAnalysis	attemptId, weakTopics[], narrative, studyPlan[], modelVersion, prompt	→ Attempt, → Result
ProctorSession	attemptId, events[], snapshots[], recordingKey, aiFlags[], reviewDecision, reviewerId	→ Attempt

Collection	Key fields	Relationships
Plan / Coupon / Campaign	pricing, interval; code, type, value, constraints, scope; utm, dates	→ Taxonomy scope, Coupon → Institute (optional)
Order / Payment / Subscription / Refund / Invoice	amounts, tax, coupon, razorpayIds, status	→ Student, → Plan, → Institute
CommissionLedgerEntry	instituteId, sourcePaymentId, type(accrual/adjustment/transfer), amount, status, settlementId	→ Institute, → Payment, → Contract
Notification / AuditLog / WebhookEvent	type, payload, actor, targetRef, readAt / hash / dedupeKey	polymorphic refs

8. API Design — Representative Endpoints

EdTech B2B2C Platform — MVP Requirements · v1.0

Confidential — prepared for external engineering (Toptal)

REST, base path /api/v1. JSON only. Auth via Authorization: Bearer <jwt> unless noted **public**. List endpoints support ?page, ?limit, ?sort, ?q and resource-specific filters. Tenant scope is enforced from the token; X-Impersonate-User is honoured only for Super Admin. This is an **indicative** surface — roughly the average level of detail expected — not the exhaustive contract; the final OpenAPI spec is a build deliverable.

8.1 Auth & Users

Method	Endpoint	Purpose
POST	/auth/register	Create account (student/parent self-serve) public
POST	/auth/login	Password login → access + refresh tokens public
POST	/auth/otp/request · /auth/otp/verify	OTP login flow public
POST	/auth/refresh · /auth/logout	Rotate / revoke tokens
POST	/auth/password/forgot · /auth/password/reset	Password recovery public
POST	/auth/2fa/setup · /auth/2fa/verify	Enrol / confirm TOTP
GET	/auth/sessions · DELETE /auth/sessions/:id	List / revoke device sessions
POST	/admin/impersonation/:userId · DELETE /admin/impersonation	Start / end impersonation (Super Admin)
GET	/users/me · PATCH /users/me	Current profile
GET / POST	/users	List / create internal users (Admin)
PATCH	/users/:id/roles · /users/:id/status	Role & status management (Super Admin)

8.2 Institutes, Contracts & Commission

Method	Endpoint	Purpose
GET / POST	/institutes	List / create Institute
GET / PATCH	/institutes/:id	Read / update Institute
POST	/institutes/:id/kyc-documents	Upload KYC doc (signed-URL flow)
POST	/institutes/:id/linked-account	Create / sync Razorpay Route linked account
GET	/institutes/:id/commitment	Committed vs enrolled vs paid, shortfall projection
GET	/institutes/:id/commission-ledger	Ledger entries + running balance

Method	Endpoint	Purpose
GET	<code>/institutes/:id/settlements</code> · <code>/settlements/:id/statement.pdf</code>	Settlement history & statement (Toptal)
GET / POST	<code>/institutes/:id/contracts</code>	List / create contract (with structured terms)
POST	<code>/contracts/:id/send-for-signature</code>	Dispatch to e-sign vendor
POST	<code>/contracts/:id/approve</code> · <code>/contracts/:id/terminate</code>	Activate / terminate (Super Admin)
GET	<code>/contracts/:id/versions</code>	Version & addendum history
POST	<code>/settlements/run</code>	Trigger consolidated settlement cycle (Super Admin)

8.3 Sales CRM, Marketing & Coupons

Method	Endpoint	Purpose
GET / POST	<code>/leads</code>	List / create lead (website form posts here <code>public</code> via captcha)
PATCH	<code>/leads/:id</code> · <code>/leads/:id/stage</code>	Update lead / move pipeline stage
POST	<code>/leads/:id/activities</code>	Log call/note/task/meeting
POST	<code>/leads/:id/convert</code>	Convert to Institute
POST	<code>/leads/:id/emails</code>	Send templated email via Brevo
GET	<code>/crm/reports/pipeline</code> · <code>/crm/reports/forecast</code>	Pipeline & forecast analytics
GET / POST	<code>/coupons</code>	List / create coupon
PATCH / DELETE	<code>/coupons/:id</code>	Update / deactivate coupon
POST	<code>/coupons/validate</code>	Validate code against cart context (used at checkout)
GET / POST	<code>/campaigns</code>	List / create campaign (UTM)
POST	<code>/marketing/broadcasts</code>	Create Brevo broadcast to a segment
GET	<code>/marketing/reports/acquisition</code>	Signups by source, coupon cost, funnel

8.4 Students & Enrollment

Method	Endpoint	Purpose
GET / POST	<code>/students</code>	List (tenant-scoped) / create student
GET / PATCH	<code>/students/:id</code>	Read / update profile
POST	<code>/students/import</code> · <code>/students/import/:jobId/status</code>	Bulk CSV import (dry-run + commit) & progress

Method	Endpoint	Purpose
GET	<code>/students/import/template.csv</code>	Download import template
PATCH	<code>/students/:id/status</code>	Suspend / expire / reactivate
POST	<code>/students/:id/parent-invite</code>	Generate parent link code
POST	<code>/parent/link</code> · GET <code>/parent/students</code>	Parent links a student / lists linked students
POST	<code>/consents</code> · GET <code>/consents?studentId=</code>	Record / list consent events (signup, proctoring, data)

8.5 Catalog, Content & Media

Method	Endpoint	Purpose
GET	<code>/taxonomy?board=&class=&subject=</code>	Fetch taxonomy tree / subtree
POST / PATCH	<code>/taxonomy/nodes</code> · <code>/taxonomy/nodes/:id</code>	Manage boards/classes/subjects/chapters/topics (Admin)
GET / POST	<code>/content-items</code>	List (filtered by topic/type/status) / create metadata
POST	<code>/content-items/:id/video-upload-url</code>	Get Bunny TUS/upload credentials
POST	<code>/content-items/:id/asset-upload-url</code>	Get S3 signed PUT for PDF/resource
POST	<code>/content-items/:id/publish</code> · <code>/unpublish</code>	Workflow transitions
POST	<code>/content-items/bulk</code>	Batch create/update from metadata file
GET	<code>/learn/home</code>	Student dashboard payload (continue watching, due tests, streak, weak topics)
GET	<code>/learn/content?subjectId=&chapterId=</code>	Student-visible content list (access-filtered)
POST	<code>/content-items/:id/playback-token</code>	Issue signed, watermarked, short-expiry playback URL
GET	<code>/content-items/:id/asset-link</code>	Issue signed PDF link (with watermark params)
POST	<code>/content-progress</code> (batch)	Upsert progress events (position, %, completion, timeSpent)
POST / DELETE	<code>/content-items/:id/bookmark</code> · POST <code>/content-items/:id/notes</code>	Bookmarks & personal notes
POST	<code>/downloads/:contentItemId/license</code>	Issue device-bound offline license (mobile, if allowed)
POST	<code>/webhooks/bunny</code>	Encode-complete / failure public + signed

8.6 Assessments, AI & Proctoring

Method	Endpoint	Purpose
GET / POST	<code>/questions</code> · <code>/questions/import</code>	Question bank CRUD & bulk import

Method	Endpoint	Purpose
GET / POST	<code>/tests</code> — MVP Requirements · v1.0	List / create test (fixed list or blueprint)
GET	<code>/tests/:id</code> · PATCH <code>/tests/:id</code>	Read / update test config
GET	<code>/learn/tests?type=topic chapter mock halfyearly fullyearly</code>	Student-available tests
POST	<code>/tests/:id/attempts</code>	Start attempt → returns shuffled, per-student served questions
PATCH	<code>/attempts/:id/responses</code>	Autosave responses (idempotent, batched)
POST	<code>/attempts/:id/submit</code>	Submit → auto-grade MCQ, queue subjective for grading & AI
GET	<code>/attempts/:id/result</code>	Score, percentile, rank, topic/chapter heatmap, time analysis
GET	<code>/attempts/:id/review</code>	Full solution review
GET	<code>/attempts/:id/analysis</code>	AI narrative + prioritised study plan with content deep links
POST	<code>/learn/adaptive-tests</code>	Generate adaptive weak-area retest for the student
GET	<code>/grading/queue</code> · POST <code>/attempts/:id/grade</code>	Teacher grading queue & rubric submission
POST	<code>/attempts/:id/proctor/session</code>	Init proctor session (identity selfie, consent, config)
POST	<code>/proctor/sessions/:id/events</code> · <code>/snapshots</code>	Stream events & upload snapshots/recording chunks
GET	<code>/proctor/review-queue</code> · POST <code>/proctor/sessions/:id/decision</code>	Reviewer dashboard & accept/flag/invalidate decision

8.7 Payments, Billing & Notifications

Method	Endpoint	Purpose
GET	<code>/plans</code>	List purchasable plans (price + coupon-aware preview) <code>public</code>
POST	<code>/checkout/quote</code>	Server-computed price: base – coupon + tax
POST	<code>/checkout/orders</code>	Create Razorpay order (one-time)
POST	<code>/checkout/subscriptions</code>	Create Razorpay subscription
POST	<code>/checkout/verify</code>	Verify Razorpay signature & finalise (fallback to webhook)

Method	Endpoint	Purpose
GET	<code>/billing/invoices</code> · <code>/billing/invoices/:id.pdf</code>	Student/parent invoice list & PDF
POST	<code>/refunds</code> · POST <code>/refunds/:id/approve</code>	Request & approve refund (with commission clawback)
GET	<code>/finance/reconciliation?from=&to=</code>	Gross/discount/tax/net/fees/refunds/commission/revenue
POST	<code>/webhooks/razorpay</code>	payment/order/subscription/refund/transfer/settlement events public + signed
POST	<code>/webhooks/esign</code>	Signature status & signed document public + signed
POST	<code>/webhooks/brevo</code>	Delivery / open / click / unsubscribe public + signed
GET	<code>/notifications</code> · PATCH <code>/notifications/:id/read</code>	In-app notification centre
POST	<code>/devices</code> · DELETE <code>/devices/:id</code>	Register / remove push token (mobile)
GET / PUT	<code>/notifications/preferences</code>	Per-category opt-in/out
GET / POST	<code>/admin/notification-templates</code>	Manage templates & per-event rules
GET	<code>/admin/audit-log</code> · <code>/admin/webhook-events</code>	Audit & webhook inspection
GET	<code>/admin/feature-flags</code> · PATCH <code>/admin/feature-flags/:key</code>	Runtime configuration

9. Third-Party Integrations

EdTech B2B2C Platform — MVP Requirements - v1.0

Confidential — prepared for external engineering (Toptal)

Service	Used for	Integration notes
Razorpay	Orders, subscriptions, checkout, refunds, GST invoicing inputs	Signed webhooks, idempotency keys, sandbox + live keys, PCI handled by Razorpay Checkout (no card data on our servers).
Razorpay Route	Commission split & settlement to Institutes	Linked accounts + KYC, transfers on payment or scheduled payouts, on-hold transfers, reversal on refund, reconciliation via transfer.* / settlement.* .
Bunny.net	Video hosting, transcoding, CDN, DRM/token auth, watermarking	Server-issued signed playback URLs, allowed-referrer lock, TUS upload, encode webhooks, per-view watermark params.
Brevo	Transactional + marketing email, contact lists, automations	API for transactional sends with templates; list/segment sync; unsubscribe + event webhooks; separate sending domains/streams.
E-signature (Leegality / Zoho Sign / Digio)	Contract signing with legal audit trail	Create envelope from template/PDF, signer routing, OTP/Aadhaar e-sign, completion webhook returns signed PDF + evidence.
Proctoring	Integrity for subjective tests	In-house browser/app event capture + periodic snapshots for MVP; pluggable AI SDK (face/object/voice) behind an interface for later upgrade.
LLM API	Test analysis narrative & study plans	Deterministic scoring first; LLM formats guidance. Model + prompt versioned per analysis; minimal PII; server-side only with rate limits & cost caps.
FCM / APNs	Mobile push	Token registration endpoint, topic/category targeting, delivery receipts.
GA4 / Plausible, Sentry	Web analytics; error & performance monitoring	Consent-gated analytics; Sentry across API, web, and mobile with release tracking.

10. Non-Functional Requirements

EdTech B2B2C Platform — MVP Requirements · v1.0

Confidential — prepared for external engineering (Toptal)

Area	Requirement
Performance	API p95 < 400 ms for read endpoints, < 800 ms for writes (excl. external calls). Video start < 3 s on 4G. Dashboard payloads < 1.5 s.
Scalability	Support peak concurrent test-taking (target: 5,000 concurrent attempts at launch, horizontally scalable). Queues absorb email/AI/grading/report spikes without blocking request path.
Availability	99.9% monthly for API & web. Graceful degradation: content & results readable if AI or email is down.
Data durability	MongoDB replica set, automated daily backups + point-in-time recovery on production, tested restore runbook.
Security	OWASP ASVS L2 target, TLS 1.2+ everywhere, secrets in a managed store, per-tenant isolation tests, rate limiting, input validation on every DTO, signed media URLs, dependency & container scanning in CI.
Privacy	India DPDP Act alignment: purpose-limited collection, consent records, data-subject export & deletion, special handling for minors (see §11).
Observability	Structured JSON logs with correlation ids, Sentry error tracking, uptime + synthetic checks, dashboards for queue depth, webhook failures, payment failures.
Accessibility	Web & mobile target WCAG 2.1 AA: keyboard nav, focus states, contrast, screen-reader labels, dynamic type on mobile.
Localisation	English only for v1; all user-facing strings externalised (i18n-ready) for later regional languages.
Browser / OS support	Web: last 2 versions of Chrome, Edge, Safari, Firefox. Mobile: iOS 15+, Android 9+.
Maintainability	TypeScript throughout, shared types package, >70% unit coverage on domain logic, e2e suite for critical flows (auth, checkout, attempt, settlement), generated OpenAPI kept in sync.
Compliance — payments	No card/PII stored server-side; GST-compliant invoicing; auditable financial ledgers.

11. Security, Privacy & Compliance

EdTech B2B2C Platform — MVP Requirements · v1.0

Confidential — prepared for external engineering (Toptal)

- **Minors:** most students are under 18. Signup for a minor is **not complete** until a linked Parent grants consent (account use, data processing, and — separately — proctoring & recording). Consent is versioned and logged (who, what version, timestamp, IP).
- **Proctoring media:** webcam snapshots/recordings and identity selfies are sensitive. Stored encrypted at rest, access limited to Teacher/Ops with audit, default **retention 90 days** then auto-purge, and excluded from general exports.
- **Data-subject rights:** endpoints/process for data export and deletion (right to erasure), with legal-hold overrides for financial records (retained per statutory requirement, typically 8 years for invoices/tax).
- **PII minimisation to AI:** only first name + performance data sent to the LLM by default; configurable; prompts logged.
- **Access control:** server-side RBAC on every route; tenant scoping guard; impersonation restricted & fully audited; 2FA mandatory for staff.
- **Secrets & keys:** managed secret store, rotation policy, no secrets in code or client bundles; separate keys per environment.
- **Transport & storage:** TLS in transit; encryption at rest for DB, object storage, and backups; signed, short-expiry URLs for all protected media.
- **Abuse protection:** rate limiting, captcha on public forms, bot protection on auth, anomaly detection on test attempts and login.
- **Legal pages:** Terms, Privacy Policy, Refund & Cancellation Policy, and a Proctoring & Consent notice published on the website and surfaced in-app at the point of consent.
- **Audit:** immutable audit log for auth, role changes, impersonation, content publish, financial actions, contract state, and proctoring decisions.

12. Infrastructure & DevOps

12.1 Hosting recommendation

Recommended: AWS. ECS Fargate (API + workers) behind an Application Load Balancer, S3 for assets, CloudFront optional for the web app, MongoDB Atlas (peered), ElastiCache or Atlas-adjacent Redis, Route 53 DNS, ACM TLS, WAF on the ALB, Secrets Manager, CloudWatch + Sentry.

Lower-ops alternative: DigitalOcean. App Platform for API/web/workers, Spaces for assets, DO Managed MongoDB + Managed Redis, DO load balancer. Simpler and cheaper to operate for a small team; fewer knobs for fine-grained scaling.

ASSUMPTION

Final choice pending client confirmation. Architecture and IaC are written to keep either option open; nothing in the application layer is provider-locked (S3-compatible storage API, standard Mongo/Redis).

12.2 Delivery pipeline

- Mono-repo (pnpm workspaces): apps/api, apps/web, apps/website, apps/mobile, packages/common (shared types, validation, enums).
- GitHub Actions: lint → type-check → unit → build → e2e (on PR) → deploy to staging on merge → manual promote to production.
- Mobile: EAS build + submit; OTA updates for JS-only changes; staged rollout & forced-update gate.
- Database migrations versioned & run in the pipeline; seed scripts for taxonomy and demo data.
- Infra via Terraform; separate state per environment; least-privilege IAM.
- Runbooks: restore-from-backup, webhook replay, key rotation, incident response.

12.3 Environments & data

- Local (Docker Compose), Staging (sandbox integrations, open to client UAT), Production (isolated, monitored).
- No production PII in lower environments; synthetic/anonymised seed data only.

13. Assumptions & Open Questions

13.1 Working assumptions (proceeding unless corrected)

- Hosting: **AWS** (DigitalOcean acceptable fallback).
- E-signature vendor: **Leegality** (India-focused); swappable behind an interface.
- Proctoring: **in-house** event + snapshot capture for MVP, AI SDK slot reserved.
- Commission basis: **net of coupon**, with refund clawback; overridable per contract.
- Settlement: **configurable per contract** (per-transaction or weekly/monthly), default monthly.
- Pricing: **both** one-time and subscription plans available.
- AI analysis: managed **LLM API** permitted, server-side, minimal PII.
- Languages: **English only** for v1.
- Institutes get their **own web login & dashboard at launch**.
- Blog: lightweight headless CMS (Sanity); static pages acceptable at launch.
- Mobile: **phone-first**, responsive tablet layout; **offline video download included in v1**.
- Parent account **must approve** minor student signup and proctoring.

13.2 Questions to confirm before/at kickoff

1. In-app purchases: are we billing students **outside** app-store IAP (web/redirect checkout) to avoid 15–30% store fees, and is that acceptable under current store policies for this content type? (Affects mobile checkout design.)
2. Exact commission slab structure & whether “paid” or “enrolled” students count toward commitment.
3. Refund windows & access-revocation rules per plan.
4. Are half-yearly / full-yearly exams scheduled centrally (fixed dates) or per Institute?
5. Subjective grading SLA & teacher capacity model.
6. Proctoring: is continuous video recording required, or are periodic snapshots sufficient for v1?
7. Content readiness: who supplies the initial video/PDF/question catalogue, in what format, and by when? (Drives the bulk-ingestion tooling scope.)
8. Do Parents need their own paid actions (e.g. pay on behalf), or view + consent only?
9. Tax registration details & invoice format sign-off.
0. Brand, domains, and subdomain scheme (app., api., marketing apex).

14. Out of Scope for v1

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Confidential — prepared for external engineering (Toptal)

- Live / synchronous classes and webinars.
- Regional-language content and UI.
- SMS / WhatsApp notifications (interface only).
- Self-serve Institute onboarding without a signed contract.
- Public marketplace, affiliate network beyond Institute referral codes.
- Advanced gamification, social feed, student-to-student messaging.
- AI-graded subjective answers (teachers grade in v1; AI-assist is P2).
- Desktop apps; offline mode on web.
- Multi-currency / international payments.
- Data-warehouse / BI stack beyond in-app reports & CSV export.

15. Suggested Delivery Phases

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Indicative sequencing for an external team; exact estimates to be provided by the delivery partner.

Everything below the P1 line is required for launch.

Milestone	Scope	Exit criteria
M0 — Foundations	Mono-repo, CI/CD, environments, auth + RBAC + impersonation, tenancy, audit log, shared types, base design system.	Users can log in per role; staging deploys automatically; impersonation audited.
M1 — B2B core	Institutes, KYC, contracts + e-sign, referral codes, Sales CRM, coupons/campaigns, marketing website shell + lead capture.	An Institute can be onboarded and put under an ACTIVE signed contract; website leads land in CRM.
M2 — Learn	Taxonomy, content management + bulk tooling, Bunny video pipeline + signed playback + watermark, PDF delivery, student enrollment + bulk import, learning web UI, progress tracking.	A student watches a watermarked lesson and reads a watermarked PDF; progress recorded; Institute Admin bulk-imports students.
M3 — Assess	Question bank + import, test builder (blueprint + shuffle), attempt runtime, auto-grading, results + heatmaps, AI analysis + adaptive retests, teacher grading queue.	Topic/chapter/mock/half-yearly/full-yearly tests run end to end; AI weak-area analysis and adaptive retest verified.
M4 — Proctor	Proctoring for subjective tests (web + event capture), snapshot pipeline, proctor review dashboard, decisions.	A subjective test is taken under proctoring, flagged events reviewed, decision recorded.
M5 — Monetise	Plans, checkout, Razorpay orders/subscriptions, webhooks, invoices, refunds, Razorpay Route commission split + settlement + statements, finance reconciliation.	A paid enrollment splits commission to the Institute, reconciles, and appears on both dashboards.
M6 — Mobile	React Native student + parent app: auth, dashboard, video (+offline), material, tests, proctoring, analysis, payments, push. Feature parity with web learning.	App store builds pass review; parity checklist signed off; forced-update works.
M7 — Harden & launch	Notifications complete, reporting complete, load & security testing, accessibility pass, content loaded, runbooks, UAT sign-off.	NFR targets met; go-live checklist complete.
—	————— P1 launch line —————	
P1.1 fast-follow	Leaderboards, scheduled reports, referral rewards, diagnostic-test onboarding polish, doubt/Q&A threads, announcements, certificates.	—
P2	Live classes, regional languages, SMS/WhatsApp, AI grading-assist, BI/warehouse, multi-currency.	—

16. Appendix

16.1 Glossary

Institute	A school or college that signs a commission contract and enrolls its students. The B2B tenant.
DIRECT tenant	Reserved pseudo-Institute for students who sign up from the website with no Institute code.
Commitment	The minimum number of students an Institute contractually agrees to enroll in a period.
Commission basis	Whether commission is computed on the gross fee or the fee net of coupon discount.
Route	Razorpay Route — split-payment / marketplace settlement to linked accounts.
Blueprint	A rule set that builds a test by sampling N of M questions per chapter/topic/difficulty.
Topic mastery	A computed strong/borderline/weak classification per topic, from attempt performance.
Adaptive retest	An auto-generated test drawn only from a student's weak topics, with adjusted difficulty.
Impersonation	Super Admin viewing/acting as another user for support, fully audited, with limits.

16.2 Third-party accounts & keys checklist

- Razorpay account (live + test), Route enabled, webhook secret, settlement bank account.
- Bunny.net Stream library, pull zone, token authentication key, webhook secret.
- Brevo account, verified sending domains (transactional + marketing), API key, webhook secret.
- E-signature vendor account, template(s), API credentials, webhook secret.
- LLM API account, key, spend cap / budget alerts.
- Firebase project (FCM) + Apple developer account (APNs key), app identifiers.
- Apple Developer + Google Play Console accounts, app listings, privacy declarations.
- Cloud account (AWS or DigitalOcean), DNS control for all domains, TLS.
- MongoDB Atlas + Redis (managed), object storage bucket/space.
- Sentry, analytics (GA4/Plausible), uptime monitor.
- Company tax details (GSTIN, PAN), invoice numbering series, legal policy copy.

16.3 Requirement ID prefixes

FR-AUTH auth/RBAC **FR-INST** institutes **FR-CTR** contracts **FR-CRM** sales **FR-MKT** marketing **FR-STU** students **FR-CNT** content **FR-VID** video **FR-PDF** material **FR-ASM** assessment **FR-AI** AI **FR-PRO** proctoring **FR-PAY** payments **FR-COM** commission **FR-NOT** notifications **FR-APP** mobile app **FR-RPT** reporting **FR-WEB** website **FR-ADM** admin